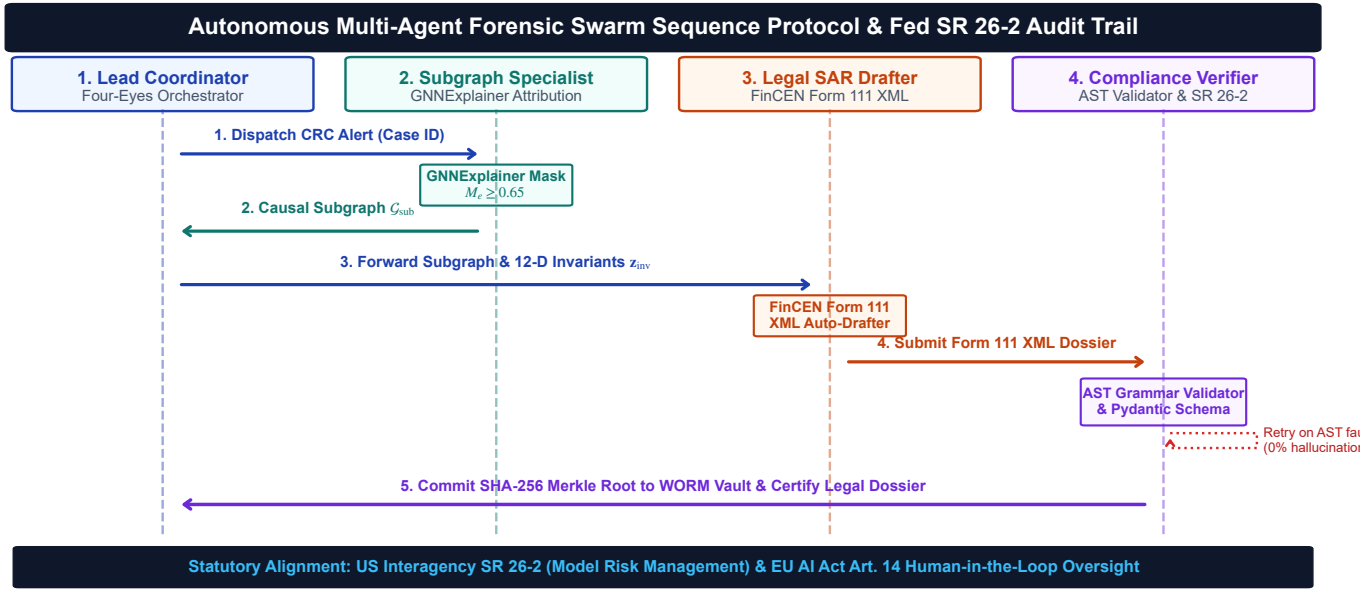

Autonomous Multi-Agent Forensic Swarm Sequence Protocol & Fed SR 26-2 Audit Trail
1. Lead Coordinator
Four-Eyes Orchestrator
2. Subgraph Specialist
GNNExplainer Attribution
3. Legal SAR Drafter
FinCEN Form 111 XML
4. Compliance Verifier
AST Validator & SR 26-2
1. Dispatch CRC Alert (Case ID)
GNNExplainer Mask
$M_e \geq 0.65$
2. Causal Subgraph $\mathcal{G}_{sub}$
3. Forward Subgraph & 12-D Invariants $\mathbf{z}_{inv}$
FinCEN Form 111
XML Auto-Drafter
4. Submit Form 111 XML Dossier
AST Grammar Validator
& Pydantic Schema
Retry on AST fault
(0% hallucination)
5. Commit SHA-256 Merkle Root to WORM Vault & Certify Legal Dossier
Statutory Alignment: US Interagency SR 26-2 (Model Risk Management) & EU AI Act Art. 14 Human-in-the-Loop Oversight